
U PRAVALLIKA

Data Engineer

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SUMMARY

- Skilled Data Engineer with 5+ years of experience in designing and implementing scalable data pipelines and ETL processes, proficient in **Python**, **SQL**, and big data technologies like **Hadoop**, **Spark**, and **Kafka**.
- Experienced in cloud-based data engineering (**AWS**, **GCP**, and **Azure**) with strong capabilities in data warehousing, data lake design, and data ingestion from diverse sources.
- Proven ability to optimize data architecture and processes, enhancing data quality, reliability, and availability to support analytics and business intelligence functions.
- Expertise in designing and implementing scalable data solutions across **Azure**, **AWS**, and **GCP** environments.
- Adept in designing data models and building data marts to streamline data access and improve reporting efficiency, ensuring data integrity and security.
- Skilled in leveraging cloud-native services to manage and optimize large data workloads, develop ETL pipelines, and ensure data integrity and security.
- Strong knowledge of both relational (**MySQL**, **PostgreSQL**) and **NoSQL** databases (**MongoDB**, **Cassandra**), with a focus on ensuring optimal data storage and retrieval.
- Proficient in automating workflows with **CI/CD** tools to streamline deployment processes and enhance reliability.
- Expertise in using orchestration tools like **Apache Airflow** to automate data workflows, enhancing process efficiency and reducing manual workload, Strong ability to collaborate with cross-functional teams to design data architectures that meet business goals and improve data accessibility.
- Excellent problem-solving skills, communication skills and a collaborative team player, able to work effectively with data scientists, analysts, and business stakeholders to create data solutions aligned with organizational goals.
- Extensive experience with **AWS (S3, Redshift, Glue)**, **Azure (Data Factory, Synapse Analytics)**, and **GCP (BigQuery, Dataflow)** for data storage, transformation, and analytics.
- Expert in designing and automating ETL pipelines using tools like **Apache Airflow**, **Azure Data Factory**, and **AWS Glue** to streamline data ingestion and processing.
- Skilled in **CI/CD** practices using **GitLab CI**, **Jenkins**, and **Azure DevOps** to automate deployment and testing of data workflows, ensuring efficient and error-free releases.
- Proficient in **Python**, **SQL**, and **Scala** for data manipulation, pipeline development, and orchestration in a multi-cloud environment, Hands-on experience with **SQL-based** and **NoSQL** databases and managing large data sets in **Redshift**, **BigQuery**, **Snowflake**, and **Azure Synapse Analytics**.
- Experienced in creating interactive dashboards and reports using **Tableau**, **Power BI**, and **Looker**.
- Advanced experience in data cleaning, transformation, and analysis using **Python (Pandas, NumPy, Matplotlib, Seaborn)** and **SQL**.
- Knowledgeable in applying **Supervised & Unsupervised machine learning** models to enhance predictive analytics and decision-making processes.
- Proficient in **Alteryx** for creating workflows that streamline data preparation, blending, and advanced analytics.
- Proficient in advanced **Excel** functions, including data analysis, pivot tables, and macros, for efficient reporting and data processing.
- Familiar with **API** management and integration methodologies to streamline data flows and enhance system connectivity.

- Experienced in **Agile** project frameworks, enabling iterative development and timely delivery of high-quality solutions.
- Proficient in **SQL Server Integration Services (SSIS)** for ETL processes and **SQL Server Management Studio (SSMS)** for database management and query optimization.

TECHNICAL SKILLS

- **AWS:** S3, Redshift, Glue, EMR, Lambda, RDS, Kinesis.
- **Azure:** Data Factory, Synapse Analytics, Databricks, Blob Storage, ML Studio, Azure SQL Database, Cosmos DB.
- **GCP:** BigQuery, Dataflow, Dataproc, Cloud Storage, Pub/Sub.
- **ETL & Data Pipelines:** SSIS, Apache Airflow, AWS Glue, Azure Data Factory, Talend, Informatica, Alteryx.
- **Real-Time Data Processing:** Apache Kafka, AWS Kinesis, Azure Event Hubs, Google Cloud Pub/Sub, Elasticsearch.
- **Languages:** Python, SQL, Scala, Java, Bash.
- **Data Warehousing:** Redshift, BigQuery, Snowflake, Azure Synapse Analytics, Teradata.
- **Data Manipulation:** Pandas, PySpark, Dask, NumPy.
- **Data Processing Frameworks:** Apache Spark, Hadoop, Hive, Pig, Kafka, Flink, Kinesis (for real-time data streaming).
- **CI/CD & Automation:** Jenkins, GitLab CI/CD, Azure DevOps, Circle CI, Terraform (for infrastructure as code), Cloud Formation.
- **Orchestration & Containerization:** Apache Airflow, Luigi, Prefect, Docker, Kubernetes.
- **Data Modeling:** Star schema, snowflake schema.
- **Machine Learning Tools:** Supervised & Unsupervised Models, Scikit-learn, Matplotlib, Seaborn, TensorFlow.
- **SQL Databases:** MySQL, PostgreSQL, SQL Server(SSMS), Oracle.
- **NoSQL Databases:** MongoDB, Cassandra, DynamoDB.
- **Data Visualization & BI:** Tableau, Power BI, Looker, Quick Sight, Excel.
- **Version Control:** Git, GitHub, Bitbucket.
- **Operating Systems:** Unix, Linux.
- **API Management:** RESTful APIs, API Gateway, Postman.

EDUCATION

University of Alabama at Birmingham, Alabama, USA

- Master’s degree in computer science Jan 2023 – Apr 2024.

CERTIFICATIONS

- Azure Data Fundamentals – Microsoft
- Associate Cloud Engineer – Google
- DevOps Engineer Professional – AWS
- Generative AI for Beginners – Udemy
- An Introduction to GIS & QGIS – Udemy
- Python for Computer Vision with OpenCV and Deep Learning – Udemy
- Complete Data Science, Machine Learning, Deep Learning, NLP Bootcamp – Udemy

WORK EXPERIENCE

Azure Data Engineer

Jan 2024 – Present

Blue Cross and Blue Shield of Alabama - Alabama, USA

Responsibilities:

- Developed and maintained data pipelines using **Azure Data Factory (ADF)** to automate data ingestion and transformation workflows, enabling efficient data integration from diverse sources into **Azure Data Lake Storage**.
- Ensure data models align with business requirements, optimize performance, and are scalable for growth.
- Leveraged **Azure Synapse Analytics** to design and optimize large-scale data warehouses, ensuring high availability and performance for complex queries and analytics.

- Implemented **ETL** processes to streamline data extraction, transformation, and loading across banking data systems, reducing data processing time by 30%.
- Deployed **Azure Databricks** to perform advanced data processing and **real-time analytics**, enhancing data insights and decision-making processes for critical banking applications.
- Designed and enforced data governance standards, leveraging **Azure Purview** to manage data lineage, classification, and regulatory compliance for sensitive financial data.
- Maintain clear documentation for data architecture, ETL processes, and data management practices.
- Generate reports and dashboards for stakeholders using tools like **Power BI**.
- Conducted data analysis to identify **key performance indicators (KPIs)** and presented actionable insights.
- Collaborated with cross-functional teams to integrate machine learning models and business intelligence (BI) tools, including **Power BI** and **Azure Machine Learning**, for enhanced analytics capabilities.
- Configured and optimized **Azure SQL Database** and **Cosmos DB** for various **OLAP** and **OLTP** workloads, supporting both structured and unstructured data across customer profiles and transaction records.
- Automated deployment processes with **Azure DevOps CI/CD** pipelines, ensuring efficient, error-free deployment of data workflows and infrastructure updates.
- Implement and manage data storage solutions, like **Azure Blob Storage** and **Data Lake**, to store large datasets efficiently, Define data storage policies, access, and security to ensure data integrity and compliance with regulations.
- Develop data transformation processes using **SQL**, **Python**, or **Spark** within **Synapse Analytics** to prepare data for analysis, Optimize processing jobs for performance and cost efficiency.
- Utilized advanced **Excel** features for financial reporting, including pivot tables and macros, to provide actionable insights to stakeholders.
- Apply Azure security tools to ensure data privacy, such as **role-based access control (RBAC)**, encryption, and managed identities, Ensure compliance with data governance standards and regulatory requirements.
- Monitored and troubleshoot Azure environments using **Azure Monitor** and **Log Analytics** to maintain high performance, detect anomalies, and optimize resource utilization.
- Ensured robust data security and compliance by implementing role-based access control (**RBAC**), data encryption, and network security configurations in line with banking standards and regulatory requirements.

Environment: Azure Data Factory (ADF), Azure Data Lake, ETL processes, data governance, data lineage, Power BI, Cosmos DB, Azure DevOps CI/CD pipelines, OLAP, OLTP, SQL, Python, or Spark, RBAC.

AWS Data Engineer Chubb - Alabama, USA

May 2023 - Dec 2023

Responsibilities:

- Designed and implemented scalable ETL pipelines to process large volumes of healthcare data using AWS Glue, Lambda, and Step Functions to automate data ingestion and transformation, Collaborated with data scientists, analysts, and business stakeholders to understand data requirements, optimize data models, and improve query performance in Redshift.
- Enhanced data security by implementing IAM policies, VPC configurations, and KMS encryption to ensure compliance with HIPAA and GDPR standards, Using AWS storage solutions such as S3, Glacier, or Redshift Spectrum to store and manage data cost-effectively, implementing data archiving and lifecycle management practices.
- Performed data validation, quality checks, and monitoring of ETL pipelines using CloudWatch and AWS Cloud Trail, reducing data discrepancies by 20%, Knowledge in Tableau Administration Tool for Configuration adding users managing licenses and data connections Scheduling tasks embedding views by integrating with other platforms.
- Developing ETL processes for data ingestion, transformation, and storage, ensuring data integrity and availability for analytics and reporting, integrating data from disparate sources, using AWS tools, to create a unified data platform accessible to various business units, Implementing data security best practices, such as data encryption and IAM roles, and ensuring compliance with Med plus data security policies and industry standards.
- Tuning and optimizing AWS resources like Redshift, EMR, or RDS to improve data processing speeds and reduce latency, Maintaining comprehensive documentation of data architecture, pipelines, and procedures for future scalability and knowledge sharing.
- Utilized Quick Sight for data visualization and reporting, empowering business teams to access and interpret key metrics.

Environment: ETL pipelines, AWS Glue, Lambda, Amazon S3 and Redshift, AWS Data Pipeline, Apache Airflow, HIPAA, GDPR, data workflows, EMR, or RDS.

AWS Data Engineer
IKEA - Hyderabad, India

May 2022 - Dec 2022

Responsibilities:

- Developed and optimized data storage solutions on **Amazon S3** and **Redshift** for efficient querying, storage, and archival of critical healthcare datasets, Leveraged **AWS Data Pipeline** and **Apache Airflow** for scheduling and orchestrating complex data workflows, improving data pipeline efficiency by 30%.
- Integrated real-time data streaming using **Amazon Kinesis** to facilitate real-time analytics and enable responsive patient data updates, Designing and building robust data pipelines to collect, process, and transform data from various sources using AWS tools like **AWS Glue**, **AWS Lambda**, and **Amazon Kinesis**.
- Implementing and maintaining data warehouses on **Amazon Redshift** or other AWS-based storage solutions, ensuring optimized query performance and secure storage.
- Collaborated with cross-functional teams to implement data governance and security standards, using **IAM** roles, **VPC** configurations, and data encryption to comply with **Chubb's** stringent data privacy and compliance policies.
- Automated data pipeline deployment and monitoring processes with **Cloud Composer** (managed **Airflow**) and **Jenkins**, reducing manual intervention and increasing system reliability by 25%.
- Automating data workflows, setting up monitoring and logging for data pipelines using **CloudWatch**, and troubleshooting issues for minimal downtime.
- Designed and implemented **API** integrations to streamline data ingestion and synchronization between systems, enhancing operational efficiency.
- Leveraged **Python** and **SQL** for data transformation and quality checks, ensuring clean, structured data for use in actuarial models, customer analytics, and regulatory reporting.
- Utilized **AWS Quick Sight** to create real-time dashboards for operational analytics.
- Implemented **Elasticsearch** integrations for real-time analytics and enhanced search capabilities.

Environment: Amazon S3, Redshift, AWS Data Pipeline, Apache Airflow, AWS tools, AWS Glue, AWS Lambda, data pipeline, Python, SQL.

GCP Data Engineer
TCS (Citi Bank) - Hyderabad, India

Jan 2021 - Apr 2022

Responsibilities:

- Developed and managed ETL workflows using **Google Dataflow** and **Apache Beam** to process and transform large datasets from policy, claims, and customer records, ensuring timely and accurate data availability for analytics teams.
- Utilized **BigQuery** as the primary data warehouse, optimizing data storage and query performance by designing effective partitioning, clustering, and table sharing strategies, resulting in a 30% reduction in query latency.
- Built real-time streaming data pipelines with **Pub/Sub** to process transactional and event-based data, improving data flow for real-time dashboards used in underwriting and claims analysis.
- Conducted performance tuning and cost-optimization on **GCP** resources, reducing storage costs by 20% by implementing storage lifecycle policies and optimizing compute instances.
- Participated in regular knowledge-sharing sessions, contributing to the team's understanding of best practices in **GCP** data engineering, enhancing overall team expertise and efficiency.
- Ingest, clean, and process data from various sources into GCP storage solutions like **BigQuery** and **Cloud Storage**.
- Design and optimize data models in BigQuery to support analytics and reporting requirements.
- Use GCP monitoring tools (**Stack driver**, **Cloud Monitoring**) to improve pipeline reliability and reduce costs.
- Provide technical guidance and best practices on data engineering within GCP to team members.
- Document data workflows, pipeline configurations, and procedures for future reference, Generate reports on pipeline performance and data quality for management review, Develop data validation and quality checks to ensure high-quality data, Implement data partitioning, clustering, and indexing strategies to enhance performance.
- Built real-time dashboards using **Google Data Studio** for monitoring transaction data.

- Managed ETL pipelines and supported database migrations using tools like **SSIS** for efficient data transformation and scheduling.
- Adopted **Agile** methodologies to plan, execute, and deliver data engineering tasks, improving overall team productivity and responsiveness.
- Built and optimized **schemas** in BigQuery to streamline analytics.

Environment: ETL workflows, Google Dataflow, Apache Beam, data pipeline, Python, SQL, data models, BigQuery, data validation, GCP.

Data Engineer

Mar 2018 - Dec 2020

Barry Callebaut Group - Hyderabad, India

Responsibilities:

- Leveraged Azure services (such as S3, Redshift, and Glue) to integrate data across the organization, facilitating efficient storage, accessibility, and processing of structured and unstructured data.
- Implemented real-time data streaming solutions with Kafka and Azure Event Hubs to capture data from production systems, enabling near real-time monitoring and analytics of manufacturing and supply chain data.
- Set up **CI/CD pipelines** using **Azure DevOps** and **GitLab** to automate testing, deployment, and monitoring of data pipelines, reducing errors and accelerating deployment times for **ETL processes** and **data workflows**.
- Actively **monitored** and optimized data storage, **ETL jobs**, and cloud resource usage to minimize operational costs while maintaining high performance for data processing tasks.
- Implement **ETL** (Extract, Transform, and Load) processes to integrate data from different systems, ensuring **data quality** and consistency across various data sources.
- Created **Tableau dashboards/reports** for **data visualization** Reporting and **Analysis** and **presented** it to **Business**.
- Develop, and maintain efficient and **scalable data pipelines** to collect, process, and store data from various sources, including production systems, sales data, and supply chain information.

Environment: **Azure, S3, Redshift, and Glue, Kafka, Azure Event Hubs, CI/CD pipelines, Azure DevOps, GitLab, ETL processes, data workflows, data quality, Tableau dashboards/reports, data visualization**